Qualitative Measurements of Policy Discrepancy for Return-Based Deep Q-Network

Wenjia Meng, Qian Zheng, Long Yang, Pengfei Li, and Gang Pan

***Abstract*— The deep Q-network (DQN) and return-based reinforcement learning are two promising algorithms proposed in recent years. The DQN brings advances to complex sequential decision problems, while return-based algorithms have advantages in making use of sample trajectories. In this brief, we propose a general framework to combine the DQN and most of the return-based reinforcement learning algorithms, named R-DQN. We show that the performance of the traditional DQN can be significantly improved by introducing return-based algorithms. In order to further improve the R-DQN, we design a strategy with two measurements to qualitatively measure the policy discrepancy. We conduct experiments on several representative tasks from the OpenAI Gym and Atari games. The state-of-the-art performance achieved by our method with this proposed strategy validates its effectiveness.**

***Index Terms*— Deep Q-network (DQN), policy discrepancy, reinforcement learning, return-based algorithm.**

I. Introduction

Reinforcement learning has achieved impressive performance on sequential decision problems [1]–[3]. The most recent successful reinforcement learning method is the deep Q-network (DQN) [3], which combines Q-learning with a deep neural network. It kick-starts recent advances in complex sequential decision-making problems. The success of the DQN largely benefits from experience replay [4], which enables it to perform the update from samples [3]. However, the traditional DQN is a bootstrap method, which only makes use of one-step samples. Such a traditional DQN is not stable enough with function approximation [5]. A promising approach to address such unstability is to combine the DQN with return-based algorithms that learn from sampled multistep returns [5]–[7].

These works [5]–[7] improve the training model by merging the DQN with a new formulation of returns. However, the formulations of returns in these works are specialized, resulting in less feasibility to make full use of different return-based algorithms [5]–[7]. Moreover, the qualitative policy discrepancy is not explicitly considered in these works, which is beneficial to utilizing off-policy returns [11]. In order to address these two issues, we propose a unified formulation that can combine the DQN and most return-based algorithms and study to qualitatively measure policy discrepancy in this unified framework. Specifically, in the proposed unified formulation, we adopt off-policy corrections to correct returns since original return-based methods are on-policy and do not work well with experience replay [5], [11]. Such off-policy corrections are based on trace coefficients, which are used to calculate the utilization level of returns [5]. These coefficients can correct policy discrepancy [5], which represents the discrepancy between target policy π and behavior policy μ. It is noticeable that π and μ represent the policy being learned about and the policy used to generate behavior, respectively [1]. In the proposed unified framework, we further study qualitative policy discrepancy, qualitative classification about whether target and behavior policies are similar or not, to enable the trace coefficient to automatically achieve a reasonable value in *near on-policy case* and *near off-policy case* (defined in Section III-B) [11]. Our contributions can be summarized as follows.

1) We propose a general framework for the R-DQN. With such a framework, most of the return-based reinforcement learning algorithms can be combined with the DQN.
2) We present a strategy with two measurements to qualitatively measure the policy discrepancy under our R-DQN framework.
3) We show, by experiments, that the performance of the existing DQN method can be significantly improved with the proposed R-DQN framework and the state-of-the-art performance can be achieved with the proposed strategy.

II. Related Work

As we focus on combining return-based algorithms and DQN in this brief, we briefly review these two components.

A. Return-Based Algorithms

Return-based algorithms are effective on estimating value function in reinforcement learning [1], [12], [27]. Original return-based methods which are on-policy are always considered as less effective on tasks with experience replay [3]. Off-policy corrections are widely used to address this problem [5], [11]. Such off-policy corrections can be regarded as an approach to combine on-policy and off-policy methods [13], [14]. The off-policy corrections in return-based algorithms depend on policy discrepancy. Most existing return-based algorithms can be divided into four categories according to the degree of dependence on policy discrepancy. In the following, we briefly introduce these return-based algorithms by category.

The return-based algorithms in the first category do not take policy discrepancy into account at all, e.g., P&W's Q(λ) [9], [15] and General Q(λ) [10]. The return-based algorithms in the second category implicitly consider policy discrepancy by correcting rewards, e.g., $Q^{\pi}(\lambda)$ and $Q^{*}(\lambda)$ [11]. The return-based algorithms in the third category explicitly consider policy discrepancy in terms of target policy, e.g., Watkins's Q(λ) [8] and TB(λ) (Tree-backup) [16]. The return-based algorithms in the fourth category explicitly take policy discrepancy into account in terms of the likelihood ratio between target policy and behavior policy, e.g., Importance Sampling (IS) [16] and Retrace(λ) [5].

Even though these return-based algorithms consider policy discrepancy in varying degrees, they do not address qualitative policy discrepancy, which can enable algorithms to efficiently benefit from

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TABLE I
SUMMARY OF RETURN-BASED ALGORITHMS

Algorithm	$Z(x')$	δ_t	C_s
Watkins's Q(λ)	$\max_a Q(x', a)$	$r_t + \gamma \max_a(Q(x_{t+1}, a)) - Q(x_t, a_t)$	λ
P & W's Q(λ)	$\max_a Q(x', a)$	$r_t + \gamma \max_a(Q(x_{t+1}, a)) - \max_a(Q(x_t, a))$	λ
General Q(λ)	$\mathbb{E}_\pi Q(x', \cdot)$	$r_t + \gamma \mathbb{E}_\pi Q(x_{t+1}, \cdot) - \mathbb{E}_\pi Q(x_t, \cdot)$	λ
IS	$\mathbb{E}_\pi Q(x', .)$	$r_t + \gamma \mathbb{E}_\pi Q(x_{t+1}, \cdot) - Q(x_t, a_t)$	$\frac{\pi(a_s\|x_s)}{\mu(a_s\|x_s)}$
TB(λ)	$\mathbb{E}_\pi Q(x', \cdot)$	$r_t + \gamma \mathbb{E}_\pi Q(x_{t+1}, \cdot) - Q(x_t, a_t)$	$\lambda \pi(a_s\|x_s)$
$Q^\pi(\lambda)$	$\mathbb{E}_\pi Q(x', \cdot)$	$r_t + \gamma \mathbb{E}_\pi Q(x_{t+1}, \cdot) - Q(x_t, a_t)$	λ
Retrace(λ)	$\mathbb{E}_\pi Q(x', \cdot)$	$r_t + \gamma \mathbb{E}_\pi Q(x_{t+1}, \cdot) - Q(x_t, a_t)$	$\lambda \min(1, \frac{\pi(a_s\|x_s)}{\mu(a_s\|x_s)})$
Our QM(λ)	-	-	$\begin{cases} \lambda \min(1, \frac{\pi(a_s\|x_s)}{\mu(a_s\|x_s)}), & \text{near on-policy} \\ \lambda \pi(a_s\|x_s), & \text{near off-policy} \end{cases}$

returns by distinguishing near on-policy case from near off-policy case [11].

B. DQN

The DQN can provide rich representations of the environment to perform well [3], [7], [17]–[19]. Many extensions have been proposed to enhance its speed or stability. Here, we introduce several representative works among these extensions.

Double DQN [20] addresses the overestimation issue of the DQN by decomposing action selection and action evaluation. The dueling network architecture [21] can generalize learning across actions to achieve better policy evaluation by utilizing the dueling network to separately represent state value function and advantage function. Prioritized experience replay [22] can replay important transitions more frequently to enable DQN to learn more efficiently. Rainbow [23] achieves the state-of-the-art results by integrating the ideas of different DQN algorithms. Safe and efficient off-policy reinforcement learning [5] can enable DQN to benefit from specific return by combining DQN with Retrace(λ). However, these works do not focus on the general return and the sample trajectory is not efficiently well used.

III. UNIFIED RETURN-BASED UPDATE TARGET

In order to combine the general return-based algorithm with DQN, we propose a unified return-based update target in this section. Based on this unified update target, we give two definitions of near on- and off-policy cases and propose a new strategy, called QM(λ), to qualitatively measure policy discrepancy.

The proposed unified return-based update target $Y(x, a)$ for the action value $Q(x, a)$ is formulated as

$$Y(x, a) = r_0 + \gamma Z(x') + \mathbb{E}_\mu \left[\sum_{t \geq 1} \gamma^t \left(\prod_{s=1}^{t} C_s \right) \delta_t \right] \tag{1}$$

where r_0 represents the immediate reward, γ is the discount rate, $Z(x')$ estimates the expectation of state value for next state x', C_s represents the nonnegative trace coefficient, and δ_t represents the temporal difference error at step t.

In the formulation of the unified return-based update target, the trace coefficient C_s is critical as it is related to the decay of trace. For near on-policy case, the trace coefficient is expected to make full use of return. For near off-policy case, the trace coefficient is expected to efficiently cut the trace. In the following, we show several typical return-based algorithms corresponding to the works introduced in Section II-A under the proposed unified formulation (1). A brief summation of these algorithms is shown in Table I. It is noticeable that the definitions of $Z(x')$, δ_t, and C_s in Table I can be found in [5] and [11].

A. Algorithm Analysis Under the Unified Formulation

No Policy Discrepancy: P&W's Q(λ) [9] and general Q(λ) [10] do not consider policy discrepancy. The computation of their trace coefficient C_s is not related to policy discrepancy.

Implicit Policy Discrepancy: Although $Q^\pi(\lambda)$ and $Q^*(\lambda)$ [11] do not consider policy discrepancy in terms of trace coefficient C_s, it implicitly takes account of policy discrepancy in terms of correcting rewards with an off-policy correction $\gamma\, \mathbb{E}_\pi\, Q(x_{t+1}, \cdot) - Q(x_t, a_t)$.

Policy Discrepancy in Terms of Target Policy: Watkins's Q(λ) [8] and TB(λ) [16] explicitly consider policy discrepancy in terms of the target policy rather than correcting rewards.

1) Watkins's Q(λ) directly cuts the trace by setting trace coefficient C_s to 0 when the sampled action is not the greedy one under target policy.
2) TB(λ) adopts trace coefficient C_s, which is proportional to target policy probability $\pi(a_s|x_s)$ to discount the trace rather than cutting it.

Policy Discrepancy in Terms of IS Ratio: IS [16] and Retrace(λ) [5] consider policy discrepancy by taking both target and behavior policies into account.

1) IS corrects policy discrepancy by setting the trace coefficient C_s to be proportional to the ratio $\frac{\pi(a_s|x_s)}{\mu(a_s|x_s)}$, where $\pi(a_s|x_s)$ and $\mu(a_s|x_s)$ represent target and behavior policies, respectively [(x_s, a_s) stands for a given state action pair].
2) Retrace(λ) allows consideration of policy discrepancy by utilizing an IS ratio truncated at 1.

B. Qualitative Policy Discrepancy: Near On- and Off-Policy Cases

Even though algorithms above consider policy discrepancy in varying degrees, they do not address qualitative policy discrepancy, which can enable the trace coefficient to achieve a reasonable value on near on- and off-policy cases [11]. Therefore, we propose a strategy that can qualitatively measure policy discrepancy, which is inspired by the formulations of the trace coefficient in the algorithms above. Algorithms with our strategy are expected to differentiate near on- and off-policy cases.

We propose two definitions for near on- and off-policy cases. These two definitions can classify whether a transition (x, a, r, x') is near on-policy case or not. Such definitions are according to two different views from the idea of the dueling architecture [21]. That is, focusing on state is more desirable when actions do not affect the environment, (i.e., when actions are irrelevant to the environment change) while focusing on action is more desirable when actions can affect the environment, (i.e., when actions are relevant to the environment change). This indicates that the effectiveness of emphasizing states depends on whether actions affect the

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environment or not. Therefore, one of our definitions is proposed to emphasize sampled states when actions do not affect the environment, while the other is proposed to emphasize the sampled action when actions affect the environment. These two definitions are detailed as follows.

Definition 1 (Near On- and Off-Policy Cases[1]*):* Given one transition, for the sampled state, when its greedy actions under target policy and behavior policy are the same, this transition is near on-policy case. Otherwise, it is near off-policy case.

Definition 2 (Near On- and Off-Policy Cases): Given one transition, for the sampled state-action pair, when the sampled action under two policies is both greedy or non-greedy, this transition is near on-policy case. Otherwise, it is near off-policy case.

With such definitions, near on- or off-policy case in return-based algorithms can be differentiated by a given bound. Once near on- or off-policy case can be differentiated, the return can be more accurately approximated by choosing a more reasonable trace coefficient. The trace coefficient C_s in the proposed strategy, QM(λ), can be expressed as

$$C_s = \begin{cases} \lambda \min\left(1, \dfrac{\pi(a_s|x_s)}{\mu(a_s|x_s)}\right), & \text{near on-policy case} \\ \lambda \pi(a_s|x_s), & \text{near off-policy case.} \end{cases}$$

Our strategy separately adopts the relatively large trace coefficient from Retrace(λ) and the relatively small one from TB(λ) for near on- and off-policy cases, i.e., $\min(1, (\pi(a_s|x_s)/\mu(a_s|x_s))) \geq \pi(a_s|x_s)$ [5]. Such setting enables the algorithm to separately make full use of returns and efficiently cut the off-policy returns on these two cases.

It should be kindly noted that $Z(x')$ and δ_t in the proposed QM(λ) vary with original return-based algorithms. As shown in Table I, these existing return-based algorithms can be classified into four categories according to the formulations of $Z(x')$ and δ_t. Each return-based algorithm category corresponds to one specific QM(λ) formulation. Specifically, QM(λ) formulations based on Watkins's Q(λ), P & W's Q(λ), and General Q(λ) are different from each other. The QM(λ) formulations based on IS, TB(λ), $Q^{\pi}(\lambda)$, and Retrace(λ) are the same. For the QM(λ) formulations based on IS, TB(λ), $Q^{\pi}(\lambda)$, and Retrace(λ), their trace coefficients $C_s \in [0, (\pi(a_s|x_s)/\mu(a_s|x_s))]$ ensure that their update targets $Y(x, a)$ in formulation (1) converge to the estimated values, i.e., the value function for a policy π (Q^{π}) and the optimal value function (Q^*) [5].

IV. R-DQN Framework With Two Measurements

As introduced in Section I, existing works [5]–[7] which combine the DQN with return-based algorithms cannot fully benefit from general return-based algorithms [e.g., Watkins's Q(λ), P&W's Q(λ), and General Q(λ)]. Therefore, in this section, we propose an R-DQN framework that can combine the DQN with the proposed unified return-based update. We first describe how to combine the DQN with the general return-based algorithms. Under this R-DQN framework, we then propose two measurements and give their bounds to qualitatively classify near on- and off-policy cases for our QM(λ). The whole algorithm of our R-DQN is summarized in Algorithm 1.

A. Proposed R-DQN

The pipeline of R-DQN algorithms is shown in Fig. 1. As shown in this figure, transitions $(x_t, a_t, r_t, x_{t+1}, \cdots, x_{t+k})$ are drawn from replay memory D. The transition sequences are utilized by the R-DQN to compute the state value estimate and temporal difference error. The loss can be formulated as

$$L(\theta_j) = (Y(x_t, a_t) - Q(x_t, a_t; \theta_j))^2$$

where θ_j represents the parameters of the R-DQN at step j. In the formulation, $Y(x_t, a_t)$ is represented as

$$Y(x_t, a_t) = r(x_t, a_t) + \gamma Z(x_{t+1}) + \sum_{s=t+1}^{t+k-1} \gamma^{s-t} \left(\prod_{i=t+1}^{s} C_i\right) \delta_s$$

where k represents the number of transitions. When updating the R-DQN, corresponding gradient descent is performed

$$\nabla_{\theta_j} L(\theta_j) = (Y(x_t, a_t) - Q(x_t, a_t; \theta_j)) \nabla_{\theta_j} Q(x_t, a_t; \theta_j).$$

Experience replay [3], [24] is adopted in the R-DQN. There are two differences between the R-DQN and the DQN in experience replay: 1) given state x, behavior policy $\mu(\cdot|x)$ is stored in the R-DQN and 2) samples drawn from replay memory in the R-DQN framework are sequential.

Algorithm 1 Return-Based Algorithm for the DQN

Require: Replay memory D with capacity N; Action-value function Q with random weights θ
Require: Target action-value function $\hat{Q}$ with weights $\theta^- = \theta$; Replacement frequency F of target network
for episode $= 1, M$ **do**
  Initialize state sequence $x_1 = \{\mathbf{s}_1\}$
  for step $j = 1, T$ **do**
    With probability ϵ select a random action a_j
    otherwise select action $a_j = \arg\max_a Q(x_j, a; \theta_j)$
    Execute a_j in emulator, observe reward r_j, state $\mathbf{s}_{j+1}$
    Set state $x_{j+1} \leftarrow \mathbf{s}_{j+1}$ and add transition tuple (x_j, a_j, r_j, x_{j+1}) to D and store behavior policy $\mu(\cdot|x_j)$ to D
    Randomly sample a minibatch of sequential transitions from D: $(x_t, a_t, r_t, \ldots x_{t+k})$ and behavior policy sequences $\mu_t, \cdots, \mu_{t+k-1}$
    Compute $Y(x_t, a_t) = r(x_t, a_t) + \gamma Z(x_{t+1}) + \sum_{s=t+1}^{t+k-1} \gamma^{s-t} (\prod_{i=t+1}^{s} C_i) \delta_s$
    Perform a gradient descent step on $(Y(x_t, a_t) - Q(x_t, a_t; \theta_j))^2$ with respect to the network parameters θ_j
    Every F steps reset $\hat{Q} = Q$
  end for
end for

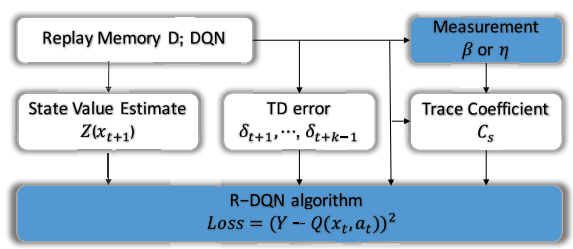

Fig. 1. Framework of the R-DQN. Given replay memory and DQN, we can obtain the proposed unified return-based update target $Y(x_t, a_t)$. This target can be used to compute the loss of return-based DQN.

B. Qualitative Measurements and Their Corresponding Bounds in R-DQN

Definitions 1 and 2 semantically define near on- and off-policy cases, but they cannot directly formulate these two cases.

[1]Researchers call it on-policy learning when target policy is the same as behavior policy, otherwise, off-policy learning [1].

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Some works [5], [11] give more intuitive expression of these two cases. More specifically, when behavior policy is similar to target policy, the case is regarded as near on-policy; otherwise, the case is near off-policy. Therefore, we give the formulated expression of near on- and off-policy cases

$$\text{case} = \begin{cases} \text{near on-policy}, & \text{measurement} < \text{bound} \\ \text{near off-policy}, & \text{measurement} >= \text{bound} \end{cases}$$

where measurement represents the dissimilarity between behavior policy and target policy, and bound is used to qualitatively classify these two cases.

In this section, inspired by the idea of 'off-policy-ness' [11], we define two measurements for policy discrepancy, namely β-based measurement and η-based measurement (as shown in Fig. 1) under our R-DQN framework. These two measurements are proposed according to the two definitions in Section III-B. As these two definitions, β-based measurement and η-based measurement are proposed for emphasizing state and sampled action in their corresponding specific environments accordingly. Motivated by the simplicity of L_1 distance to measure policy discrepancy as in some of the previous works [5], [11], we adopt such metric to derive these measurements' formulations. Their formulations are as follows:

$$\beta \stackrel{\text{def}}{=} ||\pi(.|x_t) - \mu(.|x_t)||_1 \tag{2}$$

$$\eta \stackrel{\text{def}}{=} |\pi(a_t|x_t) - \mu(a_t|x_t)| \tag{3}$$

when given the sampled state x_t and action a_t.

We then derive their corresponding bounds according to these two definitions of near on- and off-policy cases. The derivations for the bounds are related to the experimental exploration parameter ϵ in ϵ-greedy method in the DQN. During the derivations, we assume that the final experimental exploration parameter ϵ satisfies $0 < \epsilon \leq 1/2$. Such parameter assumption is consistent with representative works [3], [5], [21]–[23] (e.g., final maximal ϵ in [5] is equal to $1/2$, and in [3] and [21]–[23] satisfies $0 < \epsilon < 1/2$). The exploration parameters under target and behavior policies are separately represented as ϵ_π and ϵ_μ. It should be kindly noted that $|\epsilon_\mu - \epsilon_\pi| < 1/2$ can be derived according to the final exploration parameter assumption, i.e., $0 < \epsilon \leq 1/2$.

During the derivation of measurements' bounds, the formulations of target and behavior policies are critical. Here, such formulations are based on ϵ-greedy policy, which is a common method used in DQN algorithms [7]. In the following, we separately give these formulations for β-based and η-based measurements. Specifically, these formulations are proposed for DQN agents with discrete action space, whose action number is n. For β-based measurement: without loss of generality, on near on-policy case, we separately formulate target and behavior policies as $\pi(\cdot|x_t) = (1-\epsilon_\pi+(\epsilon_\pi/n), (\epsilon_\pi/n), \ldots, (\epsilon_\pi/n))$, $\mu(\cdot|x_t) = (1-\epsilon_\mu+(\epsilon_\mu/n), (\epsilon_\mu/n), \ldots, (\epsilon_\mu/n))$, where the greedy actions under target and behavior policies are the same one; on near off-policy case, we separately formulate target and behavior policies as $\pi(\cdot|x_t) = ((\epsilon_\pi/n), 1-\epsilon_\pi+(\epsilon_\pi/n), \ldots, (\epsilon_\pi/n))$, $\mu(\cdot|x_t) = (1-\epsilon_\mu+(\epsilon_\mu/n), (\epsilon_\mu/n), \ldots, (\epsilon_\mu/n))$, where the greedy actions under these two policies are not the same. For η-based measurement, the formulations of these two policies can be found in the following derivation process of the bound for η-based measurement.

In the following, we first derive the bound for β-based measurement by analyzing near on- and off-policy cases (Definition 1). We then derive the bound for η-based measurement by analyzing these two cases (Definition 2).

β-Based Measurement:

1) For the near on-policy case, according to (2) definition for β, we can derive

$$\beta = ||\pi(.|x_t) - \mu(.|x_t)||_1 = \left|\frac{\epsilon_\mu - \epsilon_\pi}{n}\right|(2n-2) < 1.$$

2) For the near off-policy case, let ϵ_π be greater than or equal to ϵ_μ without loss of generality. According to (2) definition for β, we can derive

$$\begin{aligned}\beta &= ||\pi(.|x_t) - \mu(.|x_t)||_1 = 2 - \epsilon_\pi\left(\frac{2}{n}\right) - \epsilon_\mu\left(\frac{2n-2}{n}\right) \\ &\geq 2 - \epsilon_\pi\left(\frac{2}{n}\right) - \epsilon_\pi\left(\frac{2n-2}{n}\right) = 2 - 2\epsilon_\pi \geq 1.\end{aligned}$$

3) When the case is near on-policy, β is less than 1. Otherwise, β is greater than or equal to 1. We conclude that the bound of β is 1 under the ϵ range assumption.

η-Based Measurement:

1) For the near on-policy case, we need to consider two situations where the sampled action under two policies is both greedy or non-greedy. In the first situation, we have

$$\begin{aligned}\eta &= \left|\left(1-\epsilon_\pi+\frac{\epsilon_\pi}{n}\right) - \left(1-\epsilon_\mu+\frac{\epsilon_\mu}{n}\right)\right| \\ &= \left|\frac{(n-1)(\epsilon_\mu-\epsilon_\pi)}{n}\right| < 1/2\end{aligned}$$

according to (3) definition for η. On the second situation, η can be represented as $|(\epsilon_\pi/n) - (\epsilon_\mu/n)|$, which is less than $1/2$.

2) For the near off-policy case, ϵ_π is greater than or equal to ϵ_μ without loss of generality. We need to consider two situations where the sampled action is greedy under target policy or behavior policy. In the first situation, we can derive

$$\begin{aligned}\left|\left(1-\epsilon_\pi+\frac{\epsilon_\pi}{n}\right) - \frac{\epsilon_\mu}{n}\right| &\geq \left|\left(1-\epsilon_\pi+\frac{\epsilon_\pi}{n}\right) - \frac{\epsilon_\pi}{n}\right| \\ &\geq |1-\epsilon_\pi| \geq 1/2\end{aligned}$$

according to (3) definition for η. On the second situation, we can derive

$$\begin{aligned}\left|\left(1-\epsilon_\mu+\frac{\epsilon_\mu}{n}\right) - \frac{\epsilon_\pi}{n}\right| &\geq \left|\left(1-\epsilon_\pi+\frac{\epsilon_\pi}{n}\right) - \frac{\epsilon_\pi}{n}\right| \\ &\geq |1-\epsilon_\pi| \geq 1/2.\end{aligned}$$

3) Based on the analysis of these two cases, we conclude that the bound of η is $1/2$ under the ϵ range assumption.

V. Experiments

In this section, we conduct three experiments to validate the effectiveness of the proposed methods. The first experiment is to compare the R-DQN algorithms in our framework with the traditional DQN. The second one is to validate the effectiveness of the proposed measurements. We compare the proposed QM(λ) with the state-of-the-art method in the third experiment. We carry out experiments on three representative tasks, i.e., *CartPole*,[2] *Copy*, and *Pixelcopter* from OpenAI Gym library [25], two classic tasks, i.e., *Mountain Car* and *Cliff Walking* in reinforcement learning, and four representative Atari games, i.e., *berzerk*, *hero*, *qbert*, and *seaquest*. Among these tasks, CartPole, Copy, and Pixelcopter are standard tasks of classic control, algorithmic, and pygame learning environment, respectively. All the results in Section V are averaged over ten independent learning trials with different random seeds [5], [26].

[2]In this brief, we adopt two configurations (CartPole-v1 and CartPole-v2) for CartPole. The maximum of episode steps are 500 and 1000, respectively.

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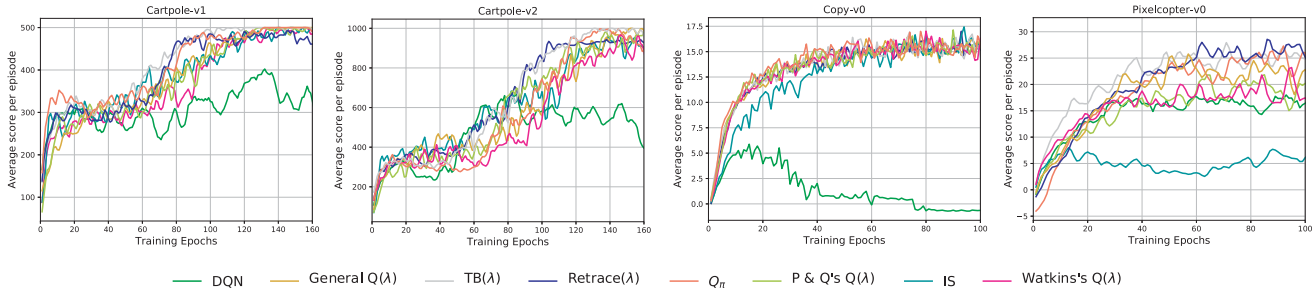

Fig. 2. Performance comparison between R-DQN algorithms and DQN. For CartPole and Copy: the network architectures are comprised of a fully connected neural network with one hidden layer of 64 neurons. For Pixelcopter: the network architecture is the same as DQN in [3]. The final exploration parameter (ϵ) for the R-DQN algorithm switches randomly (probability 0.3, 0.4, and 0.3, respectively) between the values (0.5, 0.1, and 0.01), which is consistent with that in [5]. One epoch denotes a certain amount of environment steps. For CartPole-v1, CartPole-v2, Copy, and Pixelcopter, one epoch, respectively, corresponds to 2000, 2000, 1000, and 5000 environment steps.

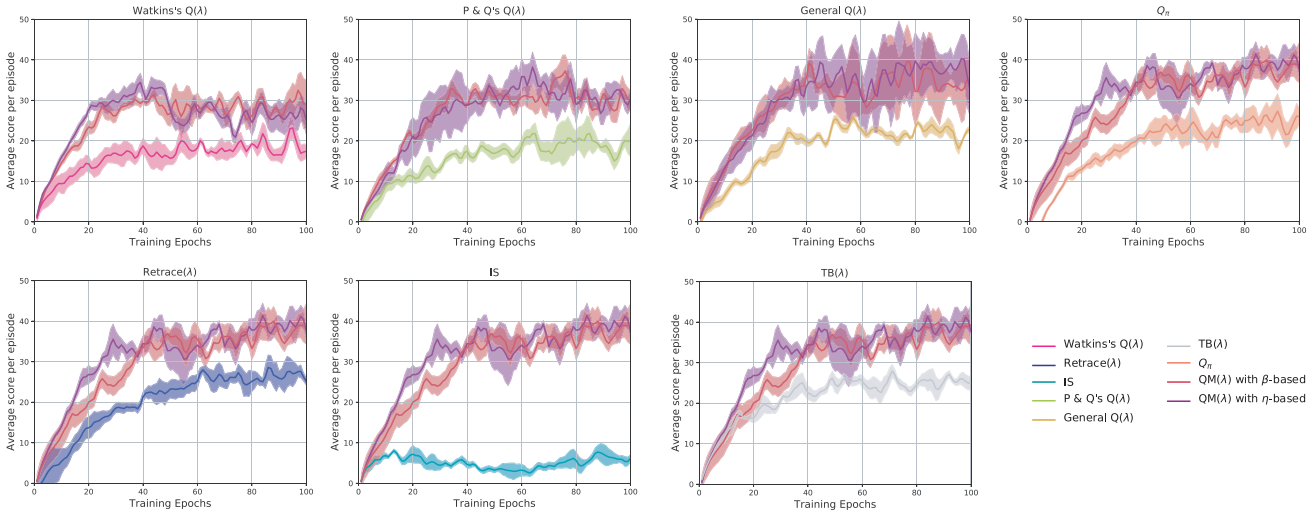

Fig. 3. Performance comparison between R-DQN algorithms with two measurements and the ones without on Pixelcopter. Bold lines: averages over ten independent learning trials. Shaded area: one standard deviation. "QM(λ) with β/η-based" is the simplified representation for the method "QM(λ) with β/η-based measurement."

TABLE II

PERFORMANCE COMPARISON AMONG ALL R-DQN ALGORITHMS ON CartPole-v1, CartPole-v2, COPY, AND PIXELCOPTER. THE REPORTED RESULTS (AVERAGED SCORE ± ONE STANDARD DEVIATION) ARE EVALUATED AT THE END OF THE TRAINING EPOCHS. "QM(λ) WITH β/η" IS THE SIMPLIFIED REPRESENTATION FOR THE METHOD "QM(λ) WITH β/η-BASED MEASUREMENT"

Methods	Cart1	Cart2	Copy	Pixel
DQN	327.3 ± 50.4	395.4 ± 93.0	-0.6 ± 0.0	16.4 ± 1.7
TB(λ)	494.7 ± 0.0	958.0 ± 15.8	14.4 ± 0.9	24.8 ± 1.5
IS	498.7 ± 19.0	905.2 ± 10.8	15.9 ± 0.4	6.1 ± 0.9
Q_π	489.9 ± 31.6	885.48 ± 55.0	15.6 ± 0.7	26.0 ± 2.0
Retrace(λ)	461.1 ± 40.7	927.4 ± 0.0	15.1 ± 0.5	25.1 ± 0.9
QM(λ) with β	**499.9 ± 3.2**	**977.0 ± 19.0**	**15.9 ± 0.2**	**39.3 ± 4.9**
QM(λ) with η	493.2 ± 12.0	947.5 ± 16.0	14.9 ± 0.1	37.5 ± 1.9
Watkins's Q(λ)	484.3 ± 0.4	887.0 ± 0.0	16.3 ± 0.1	17.4 ± 1.7
QM(λ) with β	**494.9 ± 4.2**	**908.1 ± 15.1**	16.0 ± 0.2	**29.0 ± 5.9**
QM(λ) with η	493.3 ± 1.6	904.2 ± 1.7	**16.4 ± 0.4**	25.2 ± 3.5
P & W's Q(λ)	496.7 ± 2.5	980.4 ± 74.4	15.8 ± 0.5	20.2 ± 4.1
QM(λ) with β	**500.0 ± 2.3**	**994.0 ± 15.1**	15.4 ± 0.3	29.9 ± 2.5
QM(λ) with η	499.4 ± 1.2	992.4 ± 12.1	**16.0 ± 0.4**	**30.2 ± 2.2**
General Q(λ)	499.9 ± 0.0	988.8 ± 0.0	15.2 ± 0.3	22.8 ± 0.8
QM(λ) with β	500.0 ± 0.3	**989.1 ± 4.0**	14.1 ± 0.2	**34.9 ± 6.0**
QM(λ) with η	**500.0 ± 0.2**	980.3 ± 5.2	**15.3 ± 0.1**	33.0 ± 6.5

A. Effectiveness of R-DQN Framework

We apply return-based algorithms under our R-DQN framework to the DQN to improve its performance in this section. The experimental results are reported in Fig. 2. As shown in Fig. 2, we can find that all R-DQN algorithms under our R-DQN framework achieve higher scores than the traditional DQN on tasks: CartPole-v1, CartPole-v2, and Copy. In Fig. 2, it can be observed that most R-DQN algorithms achieve higher or comparable scores compared to the DQN in Pixelcopter. Specifically, in this task, IS(λ) [16] achieves a low score as it utilizes useless trace when target policy is far away from behavior policy ($\pi(a_t|x_t) \gg \mu(a_t|x_t)$). The experimental results in Fig. 2 show that the DQN can be improved by the proposed R-DQN framework.

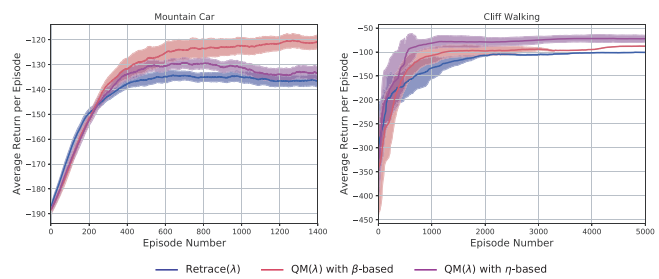

Fig. 4. Performance comparison between QM(λ) with β-based measurement and QM(λ) with η-based measurement in Mountain Car and Cliff Walking. Bold lines: averages over ten independent learning trials. Shaded area: one standard deviation.

B. Effectiveness of the Proposed Measurements

In order to evaluate the effectiveness of the proposed two measurements, we apply QM(λ) with these two measurements to R-DQN

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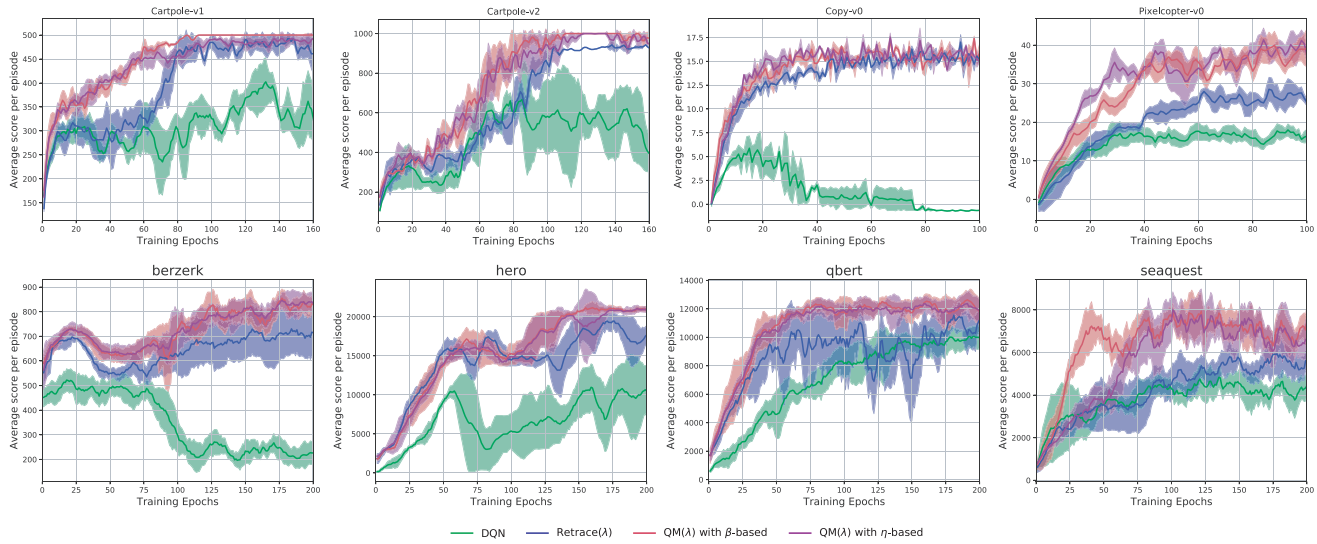

Fig. 5. Performance comparison among the DQN, Retrace(λ), and its improved algorithm with QM(λ). The performance comparison on OpenAI games and Atari games is shown in top and bottom figures, respectively. The experimental configuration on Atari games is the same as that in [3]. Bold lines: averages over ten independent learning trials. Shaded area: one standard deviation. "QM(λ) with β/η-based" is the simplified representation for the method "QM(λ) with β/η-based measurement." For these four Atari games, one epoch corresponds to 250000 environment steps.

algorithms in our framework. The experimental results are shown in Fig. 3 and Table II.[3] In the following, we separately analyze the results from Fig. 3 and Table II. In Fig. 3, it is noticeable that the combined R-DQN algorithms with these two measurements achieve higher scores than original R-DQN algorithms over the whole training period on the Pixelcopter. From Fig. 3, we can also observe that β-based measurement and η-based measurement can help R-DQN algorithms to achieve comparable scores. In Table II, R-DQN algorithms are classified into four categories according to the values of δ_t and $Z(x')$. As shown in Table II, the improved R-DQN algorithms can achieve higher scores than original R-DQN algorithms over the four categories. Such results show the effectiveness of the proposed measurements.

In order to further show the respective advantages of these two measurements, we conduct experiments on two representative samples: Mountain Car and Cliff Walking. The actions of states in Mountain Car hardly affect the environment, while the actions in Cliff Walking can make the agent fall into the cliff and return to the start position (more details about them can be found in [1]). The characteristics of these two tasks are well matched to β-based measurement and η-based measurement, respectively. The experimental results are shown in Fig. 4, and QM(λ) with β-based measurement achieves the highest average return in Mountain Car. In Cliff Walking, a gap of the QM(λ) with η-based measurement and the other algorithms can also be observed.

C. Effectiveness of the Proposed QM(λ)

In this section, we improve the state-of-the-art R-DQN method [Retrace(λ)] with the proposed QM(λ). We conduct such experiments on tasks from OpenAI Gym and Atari games, which is shown in Fig. 5. It is noticeable that the performance of the DQN is reported as a baseline in Fig. 5.

Performance on OpenAI: We validate the effectiveness of QM(λ) with two measurements on four representative OpenAI tasks, i.e., Cartpole-v1, Cartpole-v2, Copy, and Pixelcopter. The performance comparison is shown on top figures in Fig. 5. As shown in Fig. 5, R-DQNs with our QM(λ) achieve the highest scores among these compared algorithms on these OpenAI tasks. It is noticeable that R-DQNs with our QM(λ) achieve higher scores than the state-of-the-art method.

Performance on Atari: We validate the effectiveness of the proposed QM(λ) with two measurements on four representative Atari games, i.e., berzerk, hero, qbert, and seaquest. The performance comparison is reported in Fig. 5 (bottom). From Fig. 5, we can observe that our QM(λ) can help R-DQN to achieve the highest score among these compared algorithms on these Atari games. It can be noted that higher scores are achieved by the proposed QM(λ) compared with Retrace(λ).

Such experimental results validate that the proposed QM(λ) can help R-DQN outperform the state-of-the-art method.

VI. Conclusion and Future Work

In this brief, we propose an R-DQN framework. As compared to the previous works, our R-DQN framework is able to combine general return-based algorithms with the DQN. Under the R-DQN framework, we propose a strategy to reasonably benefit from off-policy returns on near on- and off-policy cases. In order to qualitatively classify these two cases, we present two qualitative measurements and further give their bounds. The experimental results show that R-DQN algorithms in our R-DQN framework outperform the traditional DQN. The effectiveness of the proposed two measurements is validated by experiments. These two measurements also show their respective advantages on different kinds of tasks. It is indicated from the results that the R-DQN with our QM(λ) can outperform the state-of-the-art method.

Limitations and Future Works: First, note that the metric of L_1 we adopt for the formulations of measurements in Section IV-B is quite simple, and it could be challenging and interesting to use other metrics to measure the policy discrepancy, e.g. Kullback–Leibler divergence. Second, despite our effects with reasonable definitions for near on- and off-policy cases in Section III-B, such definitions are limited to one-step samples. More general definitions, which can take

[3]It should be kindly noted that the results of QM(λ) based on IS, TB(λ), $Q^{\pi}(\lambda)$, and Retrace(λ) are the same due to their same QM(λ) formulations. The results of QM(λ) based on Watkins's Q(λ), P & W's Q(λ), and General Q(λ) are different due to their different QM(λ) formulations.

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multistep samples into account, are suggested as future work. Third, even though our derivation of measurement's bound in Section IV-B is well founded with a reasonable ϵ range assumption, however, it may not be applicable when there is an extreme demand for exploration due to the bounded ϵ range. Therefore, more general derivation processes are worthwhile in the future work.

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